



HWA CHONG INSTITUTION (COLLEGE SECTION)
2021 JC2 9744 H2 BIOLOGY
PRELIMINARY EXAMINATIONS PAPER 2 MARK SCHEME

QUESTION 1

- (a)(i) State the highest level of protein structure observed in Fig. 1.1. [1]
quaternary
- (ii) Describe how this structure is maintained. [3]
- ref to primary structure: each polypeptide having a specific sequence of amino acids joined by peptide bonds
 - ref to secondary structure: folding / coiling into α -helix and β -pleated sheets via hydrogen bonds
 - ref to tertiary structure: each polypeptide further bende twisted to give specific 3D conformation via R groups interactions / hydrogen bonds, ionic bonds, hydrophobic interactions
 - ref to 2 polypeptide chains with R groups interacting forming a dimer
- (iii) Explain the role of the water molecule in the active site. [2]
- ref to hydrolysis of glycosidic bonds
 - ref to sucrose breakdown
 - ref to one monomers gaining a hydrogen atom, the other a hydroxyl group
- (b) With reference to the results shown in Fig. 2.3 and Fig. 2.4, discuss which sucrose phosphorylase enzyme, free or immobilised, is better for use in industrial reactions. [4]
- immobilised
 - from pH 4.5-8, immobilised enzyme has higher activity 68-100% vs free enzyme: 32-100%
 - ref to if pH changes, less effect on immobilised enzymes ; A: more stable, pH resistant
 - ref to immobilised enzymes having higher optimum temp 75°C vs 56 °C for free enzymes
 - ref to immobilised enzymes being more active / thermostable
 - ref to advantage of immobilised: reusable / no contamination / longer shelf-life / cost effective

[Total: 10]

QUESTION 2

- (a) Describe what is meant by the term *fluid mosaic*. [2]
- ref to phospholipid molecules moving laterally / AW
 - presence of embedded protein scattered in the membrane / AW
- (b) Distinguish between these two models. [2]
- Davson-Danielli model only has proteins on the surface of the membrane while the Singer-Nicholson Model has proteins embedded within the membrane
 - Davson-Danielli model has proteins uniformly distributed across the membrane while the Singer- Nicholson Model has proteins distributed at specific locations / AW
 - Davson-Danielli model only has proteins of the same shape while the Singer- Nicholson Model has proteins of various sizes / AW

- (c) Explain how high temperature could damage the structures of cell membranes. [4]

max two from effect on phospholipids:

- high temperatures results in increased kinetic energy / disruption of hydrophobic bonds
- this causes membranes to become too fluid / disrupt integrity of membrane / more transient gaps
- ref to loss of specific membrane property (e.g. not effective as barrier / permit hydrophilic molecules through)

max two from effect on proteins:

- ref to high temperatures leading to loss of 3D conformation of proteins / proteins are denatured
- ref to disruption of protein organization / breakage of ionic, hydrogen and hydrophobic bonds
- ref to loss of specific protein function (e.g. non-functional transport protein/ enzyme activity/ cell-cell adhesion disrupted)

- (d) Suggest how the transport of testosterone into the cell would be affected by high temperatures. [2]

- as testosterone is hydrophobic and would diffuse across the membrane by passive diffusion
- high temperatures would result in increased formation of transient gaps
- thus lead to increased transport of testosterone into the cell

[Total: 10]

QUESTION 3

- (a) Describe **one** structural feature of DNA that contributes to its stability as a hereditary material. [2]

- hydrogen bonds between base pairs ;
- to stabilise the structure of the double helix / preventing DNA separation ;
- double helix / double stranded DNA ;
- protection of nitrogenous bases from degradation / resistant to spontaneous mutations ;
- each strand serving as a template to repair any DNA damage ;
- association with histone proteins ;
- folding of DNA into chromosomes to prevent damage ;

- (b) With reference to Fig. 3.1,

- (i) indicate the overall direction of replication fork in the box provided. [1]
- arrow to the left

- (ii) compare how the synthesis of lagging strand differs from the leading strand. [2]

- leading strand synthesis continuous, while lagging strand synthesis discontinuous
- leading strand synthesis towards replication fork / away from origin of replication, lagging strand synthesis away from replication fork / towards origin of replication
- leading strand synthesis does not require DNA ligase, while lagging strand synthesis requires DNA ligase to join up the different Okazaki fragments

- (iii) suggest why it is necessary for two DNA polymerase enzymes to work together in a protein complex during DNA replication. [2]

- DNA polymerase only able to synthesize from 5' → 3' direction

- antiparallel nature of DNA strands, 2 DNA polymerase to synthesize in the same direction
 - increase efficiency / prevent degradation of single stranded DNA
- (c) Explain how the end replication problem arises. [3]
- DNA polymerase only extends in a 5' → 3' direction
 - unable to fill gap after the last RNA primer is removed at end of lagging strand
 - as no upstream 3' OH
 - resulting in shorter daughter strand compared to the template strand

[Total: 10]

QUESTION 4

- (a) Explain how the cell theory applies to prokaryotes. [2]
- a cell is the smallest unit of life / living organisms are composed of one or more cells
 - bacteria are unicellular living organisms exhibiting characteristics of life
 - all cells come from pre-existing cells
 - bacteria reproduces by binary fission
- (b) Distinguish between the structure of the prokaryotic and eukaryotic genomes. [2]
- prokaryotic genome is circular while eukaryotic genome is linear
 - prokaryotic genome consists of a single chromosome, while eukaryotic genome may have multiple chromosomes
 - prokaryotic genome is bound to histone-like proteins while DNA in eukaryotic genome is bound to histone proteins
 - prokaryotic genome undergoes supercoiling while eukaryotic genome undergoes a regulated specific system of packaging
 - genes are grouped into operons in prokaryotes while genes are under the control of individual promoters in eukaryotes
- (c) Describe the structural features of the mRNA transcribed from the *lac* operon. [2]
- polycistronic mRNA
 - multiple start and stop codons / Shine-Dalgarno sequences
 - no 5' cap and no 3' poly A tail
 - no introns
- (d) Using a tick (✓) or a cross (✗), indicate the presence or absence of each of the following metabolites in culture medium **A**. [1]

metabolite	presence or absence of metabolite
glucose	✗
lactose	✓

- (e) State **and** explain how the composition of culture medium **B** results in the basal expression of the *lac* operon [4]
- presence of both glucose and lactose
 - glucose decreases cAMP concentration, thus no cAMP-CAP complex formed and bound to CAP binding site
 - allolactose will bind and inactivate lac repressor which cannot bind to operator
 - RNA polymerase can bind to promoter, but not stable resulting in basal expression

[Total: 10]

QUESTION 5

- (a) Explain the role of the luciferase gene in this investigation. [2]
- to indicate if the gene expression has occurred/ promoter has been active
 - allows the level of transcription to be quantified
- (b) Comment on how these results illustrate the regulation of luciferase gene expression. [6]
- without enhancer, expression level 1x
 - with enhancer, expression level increased from 1x to 10-100x
 - activators bind to enhancer, and ref to bringing the enhancer close to the promoter
 - Improved recruitment of gTFs and RNA polymerase to the promoter, forming stable TIC
 - with silencer and enhancer, expression level $\leq 1x$
 - repressors bind to silencer, and ref to one mechanism of repression
 - activator unable to bind to the enhancer / gTFs unable to form TIC / prevent recruitment of gTF and RNA polymerase to the promoter
- (c) Discuss the significance of the specific combinations of control elements and transcription factors in regulating gene expression. [3]
- for temporal control
 - for spatial control
 - for correct numbers of proteins produced

[Total: 11]

QUESTION 6

- (a) Identify and explain to which group of genes K-ras belong to in regulating cell division. [3]
- proto-oncogenes usually encode gene products that promote normal cell division
 - ref to description of signal transduction step involving G protein
 - gain in function mutation in K-ras proto-oncogene to form oncogene
- (b) Discuss how a genetic change to the K-ras gene leads to cell proliferation. [4]
- substitution / deletion / insertion mutation of single base pairs
 - leading to missense / nonsense / frameshift mutation
 - Due to change in specific 3D conformation, KRAS is non-functional / hyperactive / excess
 - ref to *details* pertaining to the consequence of the mutation on signal transduction
- (c) Suggest why pancreatic cancer remains a challenge to be diagnose early. [2]
- ref to lack of symptoms in the early stages of pancreatic cancer
 - ref to symptoms being common to other illnesses
 - AVP

[Total: 9]

QUESTION 7

(a)(i) State Mendel's Second Law of Independent Assortment. [1]
ref to independent segregation of different allele pairs

(ii) State what is meant by *pure-breeding* in this context. [1]
homozygous at both gene loci

(b) Draw a genetic diagram to show the expected phenotypic ratio for the progeny resulting from the series of crosses between the F₁ offspring and brown, chinchilla rabbits if inheritance is Mendelian. [5]

F1 test cross black, full colour x brown, chinchilla

BbHh x **bbhh** ;

Gametes

BH **Bh** **bH** **bh** **bh** ;

Random fertilization

		Female gametes			
		BH	Bh	bH	bh
Male gametes	bh	BbHh	Bbhh	bbHh	bbhh ;

expected progeny genotypic ratio 1 **BbHh** : 1 **Bbhh** : 1 **bbHh** : 1 **bbhh** ;

expected progeny phenotypic ratio 1 black, full colour : 1 black, chinchilla : 1 brown, full colour : 1 brown, chinchilla ;

(c) Explain how it is then possible to obtain the observed numbers in the 100 progeny from the series of crosses between the F₁ offspring and brown, chinchilla rabbits. [3]

- ref to incomplete linkage
- ref to idea of crossing over
- ref to idea of production of recombinant gametes and phenotypes

(d) Suggest how researchers can use similar breeding experiments with many different pairs of characters to map the position of genes on the chromosomes of rabbits. [3]

- ref to calculation of RF / COV
- ref to idea of using RF / COV to determine distance between genes
- ref to idea of crossing over being proportional to distance between genes
- ref to analyzing relative distance between different pairs of genes

[Total: 13]

QUESTION 8

- (a) State two structural features of nucleotides that are visible in coenzyme A. [2]
- phosphates
 - pentose
 - purine
- (b) Describe the role of coenzyme A (coA) in cellular respiration. [2]
- combine with acetyl group in link reaction
 - deliver acetyl group to the Krebs cycle
- (c) Outline how ATP is synthesised in Krebs cycle. [2]
- substrate level phosphorylation
 - transfer of phosphate group from a substrate to ADP
- (d) With reference to Fig. 8.2,
- (i) explain how fluoroacetate inhibits the Krebs cycle. [3]
- fluoroacetate combine with oxaloacetate to form fluorocitrate
 - fluorocitrate is structurally similar to citrate
 - fluorocitrate is a competitive inhibitor
- (ii) suggest how the inhibition can be overcome. [1]
- addition of large amounts of citrate
- (e) Explain why the administration of fluoroacetate resulted in no ATP synthesis. [3]
- no substrate level phosphorylation
 - no reduced NAD / FAD to donate electrons to ETC
 - no oxidative phosphorylation

[Total: 13]

QUESTION 9

- (a)(i) Describe **two** differences in the evolutionary relationships amongst these five species of Pacific Salmon based on the two classifications shown in Fig. 9.1. [2]

Classification X	Classification Y
1 Pink and Sockeye share a more recent common ancestor than Chum	Pink and Chum share a more recent common ancestor than Sockeye
2 Chinook shares a more recent common ancestor with Chum, Pink & Sockeye than with Coho	Chinook shares a more recent common ancestor with Coho than with the other 3 species

- (ii) State **one** reason why classification Y is a better representation of evolutionary relationships than classification X. [1]
- all the fish possess nucleic acids
 - genome sequences facilitates the objective assessment of evolutionary relationships
 - fishes have certain molecular traits in common that offer a valid basis for comparison

(b) Complete Table 9.1 to show the classification of coho salmon.

[2]

Table 9.1

Taxon	Coho salmon
Kingdom	Animalia
Phylum	Chordata
Class	Actinopterygii
Order	Salmoniformes
Family	Salmonidae
Genus	<i>Oncorhynchus</i>
Species	<i>Oncorhynchus kisutch</i>

(c) Suggest why jacks are able to spawn as successfully as hooknoses.

[1]

- jacks are smaller and hence can hide from predators
- jacks spend less time in ocean and hence have higher chances of escaping predators
- jacks lack large teeth and hence have different source of food

(d)(i) Sketch a curve on Fig. 9.3 to show the new distribution of body length in the present-day population of reproductively mature male salmon.

[1]

Two peaks either side of normal peak

(ii) Explain the type of selection that is occurring.

[2]

- disruptive selection
- selection pressure is exerted by breeding environment
- jacks are smaller and are able to sneak better around boulders in stream bed, hooknoses are larger and are able to fight more aggressively

(e) Explain why the loss of sub-populations of the Interior Fraser population represents a decrease in biodiversity.

[3]

- decrease in genetic variation
- loss of alleles
- ref to sub-populations different due to different mutations

[Total: 13]